

PRACTICAL NO 9

Aim: Write a program to implement BFS and DFS of Binary Search Tree.

INPUT:

```
def bfs(graph, start):  
    visited = []  
    queue = []  
    visited.append(start)  
    queue.append(start)  
    while queue:  
        vertex = queue.pop(0)  
        print(vertex, end=" ")  
        for neighbour in graph[vertex]:  
            if neighbour not in visited:  
                visited.append(neighbour)  
                queue.append(neighbour)  
  
graph = {  
    0: [1, 2],  
    1: [2],  
    2: [3],  
    3: [1, 2]  
}  
print("\nBreadth First Traversal:")  
bfs(graph, 0)  
  
def dfs(graph, start):  
    visited = []  
    stack = []  
    stack.append(start)  
    while stack:  
        vertex = stack.pop()  
        if vertex not in visited:  
            visited.append(vertex)  
            print(vertex, end=" ")  
            for neighbour in graph[vertex]:  
                if neighbour not in visited:  
                    stack.append(neighbour)  
  
graph = {  
    0: [1, 2],  
    1: [2],  
    2: [3],  
    3: [1, 2]  
}  
print("\nDepth First Traversal:")  
dfs(graph, 0)
```

OUTPUT:

Breadth First Traversal:

0 1 2 3

Depth First Traversal:

0 2 3 1

